Practice Exam - 1

Question 68 of 117

A 29-year-old patient is status post orthotopic heart transplant with an anaphylactic reaction to sulfonamide antibiotics. He has a past medical history of severe persistent asthma, hypertension, hyperlipidemia, and severe G6PD deficiency. All renal and hepatic function are within normal limits.

Which one of the following would be the best therapeutic option for primary *Pneumocystis jirovecii* pneumonia prophylaxis in this patient?

A. Dapsone

B. Sulfamethoxazole/trimethoprim

C. Pentamidine

D. Atovaquone

Commentary | References

Testing point: The Board-certified AHFTC specialist should know how to prevent potential opportunistic infections post-heart transplant.

Correct Answer: D. Atovaquone

Rationale: Pneumocystis jirovecii (PJP) is an opportunistic fungal pathogen that can cause pneumonia in up to 15% of transplant recipients during periods of high dose steroid use in the absence of appropriate antimicrobial prophylaxis. The drug of choice for PJP prophylaxis is trimethoprim-sulfamethoxazole for a duration of 6 to 12 months, but its utility can be limited by allergic reactions present in approximately 3% of the population. In patients with trimethoprim-sulfamethoxazole intolerance or allergy, dapsone is often used for PJP prophylaxis. Before initiation, glucose-6-phosphate dehydrogenase (G6PD) function should be checked as a deficiency increases the risk of hemolytic anemia and methemoglobinemia. In this patient with known significant G6PD deficiency Answer A is therefore not the best choice. Sulfamethoxazole is a sulfonamide antibiotic and cannot be used in this patient with prior anaphylactic reaction to sulfonamide antibiotics; sulfamethoxazole/trimethoprim

(Answer B) should be avoided. Monthly pentamidine (Answer C) is a reasonable alternative; however, significant bronchospasm (15%) occurs with administration and can worsen symptoms in patients with underlying asthma. While prophylactic inhaled albuterol can be administered to prevent bronchospasm, this patient has severe persistent asthma, making inhaled pentamidine (Answer C) not an ideal therapeutic option. Atovaquone will adequately cover against PJP, does not contain a sulfonamide moiety and can safely be used in patients allergic to "sulfa" antibiotics, thus making Answer D the best therapeutic choice.